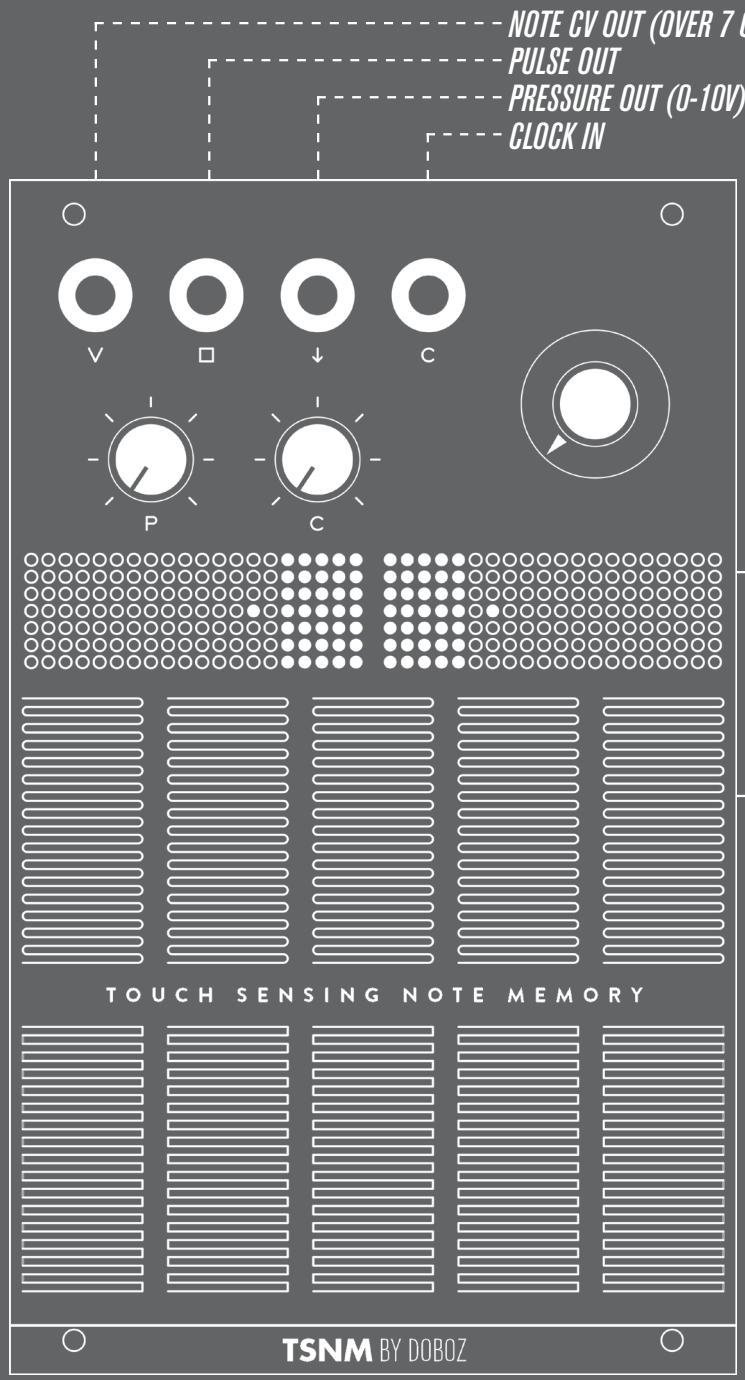
TSNM4U

TOUCH SENSING NOTE MEMORY

BY DOBOZ



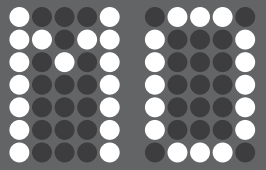
"P" PORTAMENTO POT
SETS THE SPEED OF GLIDING

"C" CLOCK POT
SETS THE RATE OF THE INTERNAL CLOCK. WHEN TURNED FULLY COUNTER-CLOCKWISE, THE INTERNAL CLOCK IS DISABLED AND THE UNIT EXPECTS A CLOCK SIGNAL ON ITS CLOCK INPUT.

ENCODER
HOLD FOR 1 SECOND TO ENTER/EXIT SETTINGS, IN ARP MODE PRESS TO LATCH ARPEGGIATOR, IN SEQ MODE PRESS TO RESET SEQUENCE.

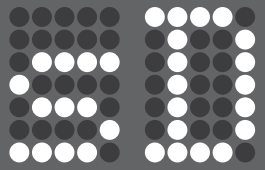
TRANSPOSE PLATES
TO EDIT TRANSPOSITION DISTANCE HOLD THE PLATE AND ROTATE ENCODER (RANGES FROM -24 TO +24 SEMITONES)

NOTE PLATES
HOLD A NOTE PLATE AND ROTATE THE ENCODER TO TUNE ITS CORRESPONDING CV. (TUNING IS QUANTISED TO SEMITONES, ALTERS CV OUTPUT OVER 3 OCTAVES)



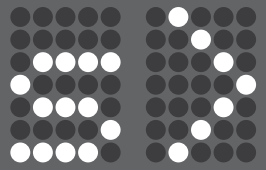
MODE

- KEYBOARD
- ARPEGGIATOR
- SEQUENCER



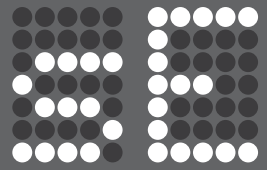
SEQ DIRECTION

- UP
- DOWN
- PING-PONG
- RANDOM



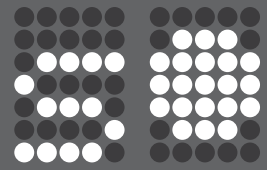
SEQUENCE LENGTH

Determines the length of the sequence, from 1 to 32 steps.



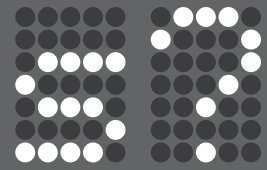
SEQUENCE EDITOR

Rotate encoder to scroll through steps. Hold left transpose plate and rotate encoder to edit note per step. Hold right transpose plate and rotate encoder to edit trigger per step. Hold both transpose plates and rotate encoder to edit length per step (L1-L8).



SEQUENCE LEARN

Touch note plates to enter the sequence. Press left transpose plate to alter step length. Press right transpose plate to toggle on/off trigger per step. Rotate encoder to undo step (no redo!). When you're done press encoder to exit learn mode.

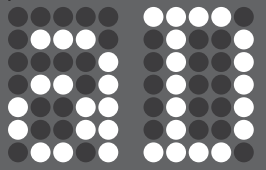


RANDOM NOTE

Hold right transpose plate and rotate encoder to edit scope of random notes (1-30). Hold left transpose plate and rotate encoder to edit probability of randomness (0-35).

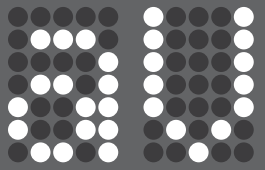
PRESENT ONLY IN ARPEGGIATOR MODE

PRESENT ONLY IN SEQUENCER MODE



ARP DIRECTION

- UP
- DOWN
- PING-PONG
- RANDOM

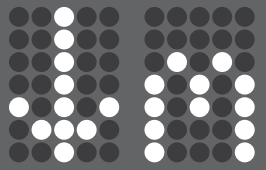


ARP VARIATOR

Hold left transpose plate and rotate encoder to choose when variation happenes:

- NEVER
- AFTER EVERY STEP
- AT THE END OF ARP SEQUENCE

Hold right transpose plate and rotate encoder to determine the number of additional notes (1-6). Hold both transpose plates and rotate encoder to determine the distance of additional notes. From the previous note (1-12 semitones).



PRESSURE: MODE

- PRESSURE
- RANDOM

Generates random voltage with every triggered note or clock pulse.

- GATE
- PULSE
- RECORD

Records and replays pressure activity up to 4 seconds. Press and hold any note plate for the desired time to set the length of recording. Hold right transpose plate and rotate encoder to set play mode:

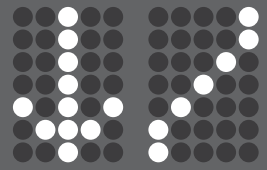
- LOOPING
- ONE-SHOT

Hold left transpose plate and rotate encoder to select reset source, which restarts the playback from the begining:

- FREE RUNNER
- TOUCH PLATES
- CLOCK

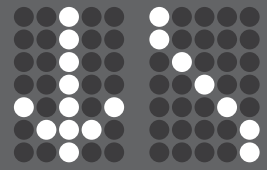
After exiting settings the loop will be overdubbed whenever a note plate is touched. To avoid this the recorded loop can be locked. Rotate encoder CW to disable overdubbing (lock loop), CCW to enable it (unlock).

- LOCKED
- UNLOCKED



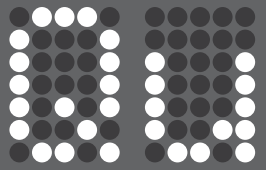
PRESSURE RISE TIME

Creates slew limiting effect on rising edge of pressure output (0-20) in all pressure modes except pulse.



PRESSURE FALL TIME

Creates slew limiting effect on falling edge of pressure output (0-20) in all pressure modes.



QUANTISER

Applies a quantification to the outgoing control voltage, which can be quantized to semitones, or to one of the scales listed below:

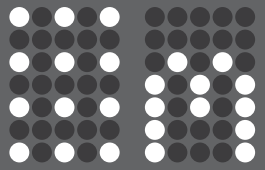
semitones, ionian, dorian, phrygian, lydian, mixolydian, aeolian, locrian, b+ blues major, b- blues minor, p+pentatonic major, p- pentatonic minor, folk, japanese, gamelan, gypsy, arabian, flamenco, whole note, quarter notes

FILL SCALE

Hold left transposition plate and press the encoder for 1 second to fill the note plates with notes of the selected scale.

C# ROOT NOTE

Hold right transposition plate and rotate encoder to set the root note.



TRANSPOSE MODE

Sets the behavior and response of transposition plates.

LA LATCHING

LI LATCHING IMMEDIATE

MO MOMENTARY

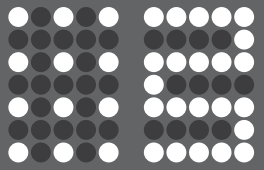
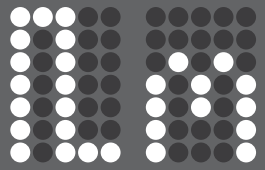


PLATE VALUES

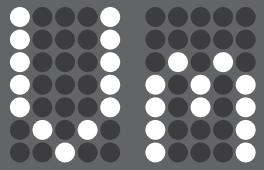
Touch and hold plates and rotate encoder to set their value. Transposition distance ranges from -24 to +24 semitones. The tuning of note plates is quantised to semitones over 3 octaves.



PULSE MODE

TR TRANSIENT (TRIGGER)

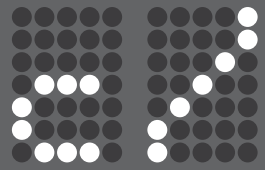
SU SUSTAINED (GATE)



VOLTAGE MODE

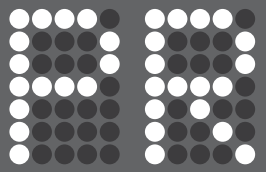
1U 1.2V/oct

2U 1V/oct



CLOCK DIVIDER

Applies division to the incoming clock signal from /1 to /32.



PRESETS

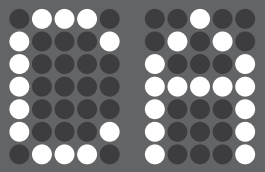
The module can save 19 presets, which includes note layout and all the parameters under settings. At startup the last recalled preset is loaded. Rotate encoder to scroll through presets.

L?

Hold the left transpose plate and press the encoder for 1 second to load the selected preset.

S?

Hold the left transpose plate and press the encoder for 1 second to save the selected preset.



CALIBRATION

To enter calibration page press and hold encoder for 1 sec. Connect your multimeter to ***NOTE CV OUTPUT***. Scroll through octave values with the encoder, to modify the value press and hold any transposition plate and rotate the encoder. Calibration must to be done for 1.2V/oct and 1V/oct modes separately.

Press and hold encoder for 1 sec to...

S? SAVE? ... save modified calibration values.

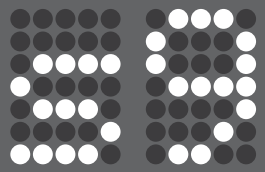
I? INIT? ... load initial calibration values.

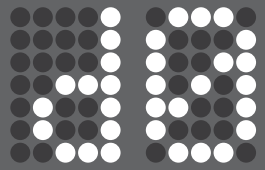
E? EXIT? ... exit calibration without saving modified calibration values.

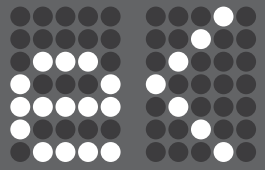
PB PRESSURE BOTTOM

PT PRESSURE TOP

Patch ***PRESSURE OUTPUT*** to control a parameter in your system. Scroll to PRESSURE BOTTOM/TOP, put your finger on a note plate and by laying more of your finger down on the touchplate, altering the pressure force listen to changes in sound controlled by the ***PRESSURE OUTPUT***. Press and hold any transposition plate and rotate the encoder to change the above mentioned values to alter the response of pressure sensing. Suggested initial values are 17 for PRESSURE BOTTOM and 65 for PRESSURE TOP.







SENSITIVITY & DEBOUNCE

Some MPR121 breakout boards may behave differently, regarding the sensitivity of touch plates. If your module in keyboard mode behaves like a totally unpredictable sequencer clocked by a Geiger counter try to lower SENSITIVITY and/or increase DEBOUNCE.

To adjust these parameters hold the encoder during power-on, then rotate it to modify their value. Press the encoder to scroll through parameters. Try to keep SENSITIVITY as high as possible and DEBOUNCE as low as possible.

The third parameter determines encoder direction. After setting these values the module needs to be rebooted by turning off then on your system.

CLARIFICATION

ARPEGGIATOR

ARPEGGIATOR WILL STEP THROUGH TOUCHED PLATES IN ORDER OF THEIR NUMBER (AS SHOWN IN THE FIGURE). THIS MEANS THAT ARPEGGIATION WON'T NECESSARY HAPPEN IN THE CLASSIC WAY, SINCE NOTE PLATES CAN BE CONFIGURED TO ANY NOTE. AFTER EXITING SETTINGS PRESSING ENCODER SWITCH WILL ACTIVATE 'HOLD' FUNCTION.

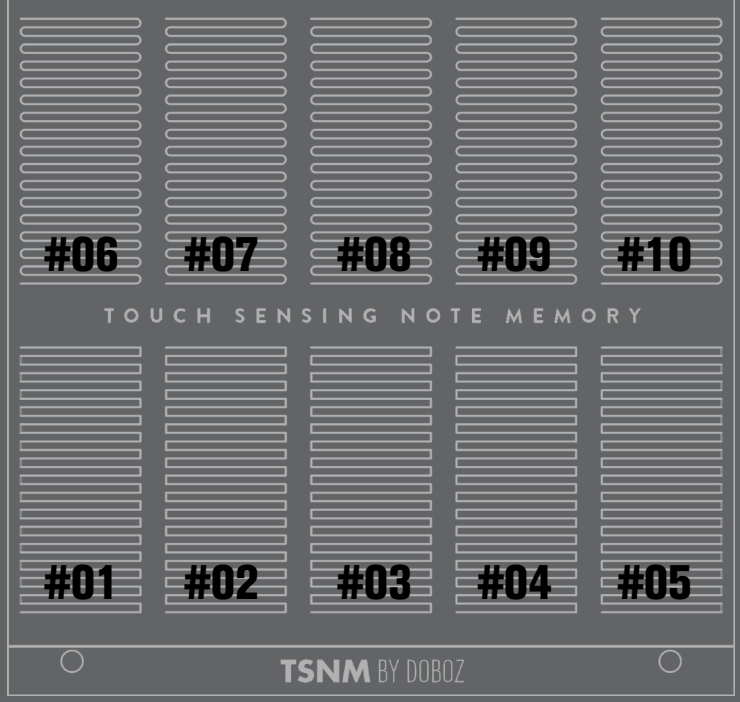
ARPEGGIATOR VARIATOR

ADDS ADDITIONAL NOTES TO ARPEGGIO. IN THE EXAMPLE: TSNM ADDS TWO MORE NOTES TO ARPEGGIATED SEQUENCE, ADDITIONAL NOTES WILL BE 5 SEMITONES HIGHER COMPARED TO THE PREVIOUS.

'x2' ARPEGGIO WITHOUT VARIATOR:
C1 - E1 - G1 ...
'!2' ARPEGGIO WITH VARIATOR:
C1 - F1 - A#1 - E1 - A1 - D2 - G1 - C2 - F2 ...
'>2' WITH VARIATIONS AT THE END OF ARPEGGIATED SEQUENCE:
C1 - E1 - G1 - F1 - A1 - C2 - A#1 - D2 - F2 ...

SEQUENCER

PLAYS A SEQUENCE OF PROGRAMMED NOTES WHICH IS TRANSPOSED BY NOTE PLATES. THE RESULT CAN BE QUANTISED. AFTER EXITING SETTINGS PRESSING ENCODER SWITCH WILL RESET THE SEQUENCE TO THE FIRST STEP.





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